

H801MPWD Board

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The H801MPWD board is an AC power board, consisting of an AC power unit and a monitoring unit. The board supplies power to the device and supports the lead-acid battery for power backup. The AC power unit provides overcurrent protection, output overvoltage protection, short circuit protection, and over-temperature protection. The monitoring unit provides comprehensive management for the power system and storage battery, and also provides certain environment monitoring functions.

Feature and Specifications

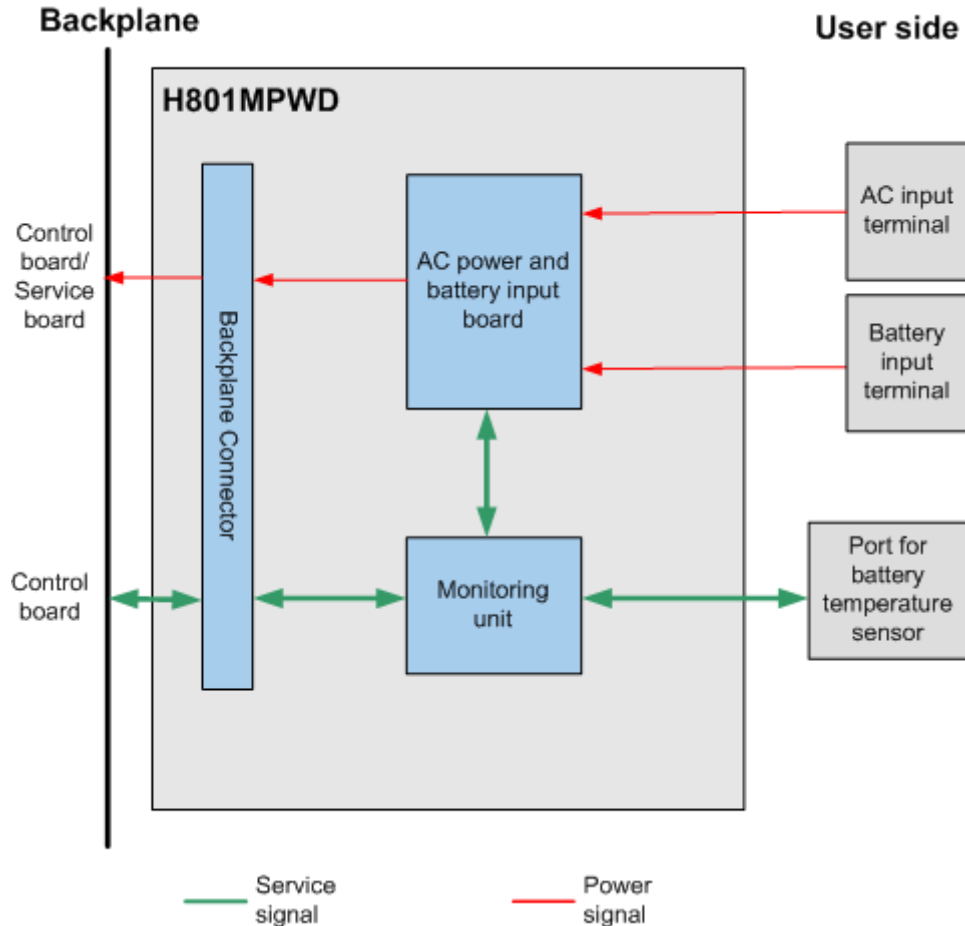
The H801MPWD board supports the following functions and specifications:

- Input AC voltage: -100 V to -240 V
- Input AC frequency: 50 Hz to 60 Hz
- Single-phase three-wire AC power input
- Battery port, a 3.3 V terminal socket, supporting DC input power
- Lead-acid battery input, reverse polarity protection, and fuse protection
- Output DC power voltage: -48 V (adjustable from -42 V to -57.6 V)
- Rated output power: 400 W
- Alarm monitoring
- Enabling and disabling of the rectifier module; equalized and float charging for the rectifier module
- Battery low voltage disconnection; charge and discharge management of storage batteries

Parameters

Please refer to "[Power Consumption and Maximum Frame Size of Boards](#)".

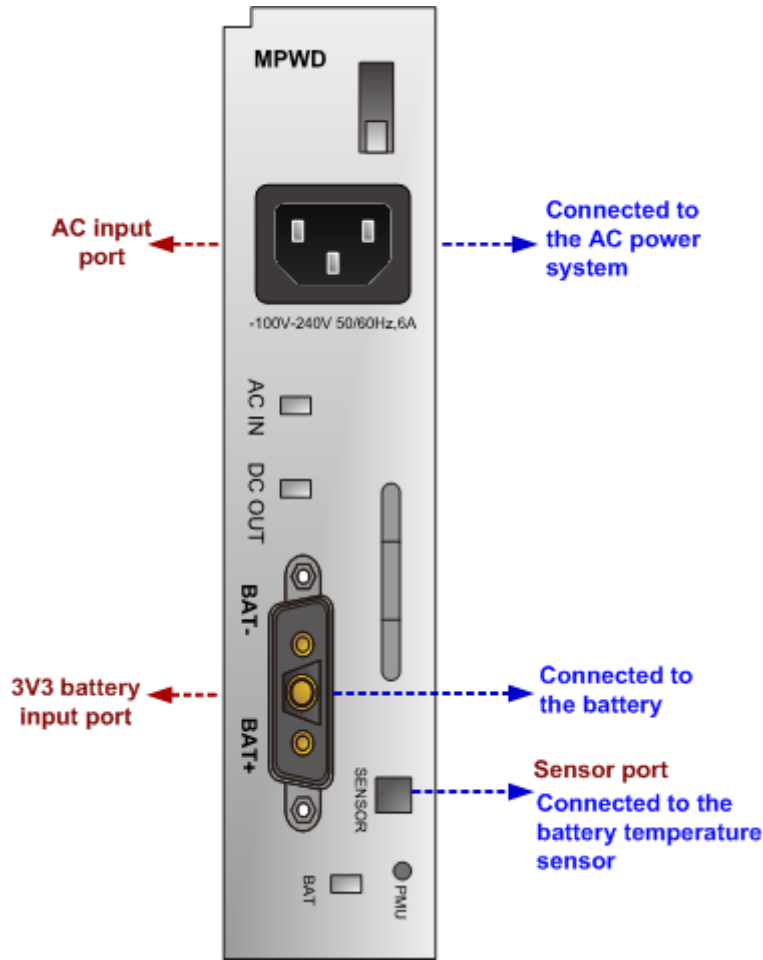
Working Principle

Figure 1 Working principle of the H801MPWD board

The basic working principle of the H801MPWD board is as follows:

- AC input power is supplied to the board. After AC/DC conversion, the power is supplied to the system backplane at -48 V (adjustable).
- The board supports battery input. The monitoring unit manages the charging and discharging of batteries.
- The board communicates with the system through the backplane, reporting board hardware and electronic label information to the backplane.
- The board supports communication with the monitoring unit through the 485 protocol to adjust voltage and current and report alarm information.
- The board supplies 3.3 V power from the system to the monitoring unit.
- The monitoring unit manages lead-acid batteries.
- The optical coupler implements the battery low voltage disconnection function.
- The board provides power distribution checking and alarm reporting functions.

Front Panel Ports



Indicator

Indicator	Name	Status	Meaning
AC IN	AC input indicator	Green: on	Indicates that the AC input power is normal
		Off	Indicates that the AC input power is abnormal
DC OUT	DC output indicator	Green: on	Indicates that the DC output power is normal
		Off	Indicates that the DC output power is abnormal
BAT	Power input indicator	Green: on	Indicates that the relay is picked up
		Off	Indicates that the relay drops out
PMU	Running status indicator	Green: blinks every 0.25s	Indicates that program loading is in progress
		Green: blinks every 1s	Indicates that the board runs normally
		Red: blinks	Indicating that the board is starting up
		Red: on	Indicates that a critical alarm is reported

Indicator	Name	Status	Meaning
		Orange: blinks	Indicates that a minor alarm (such as the high temperature alarm) is reported

Parent topic: [Power Board \(MPW\)](#).